Artificial Intelligence

CSC 401

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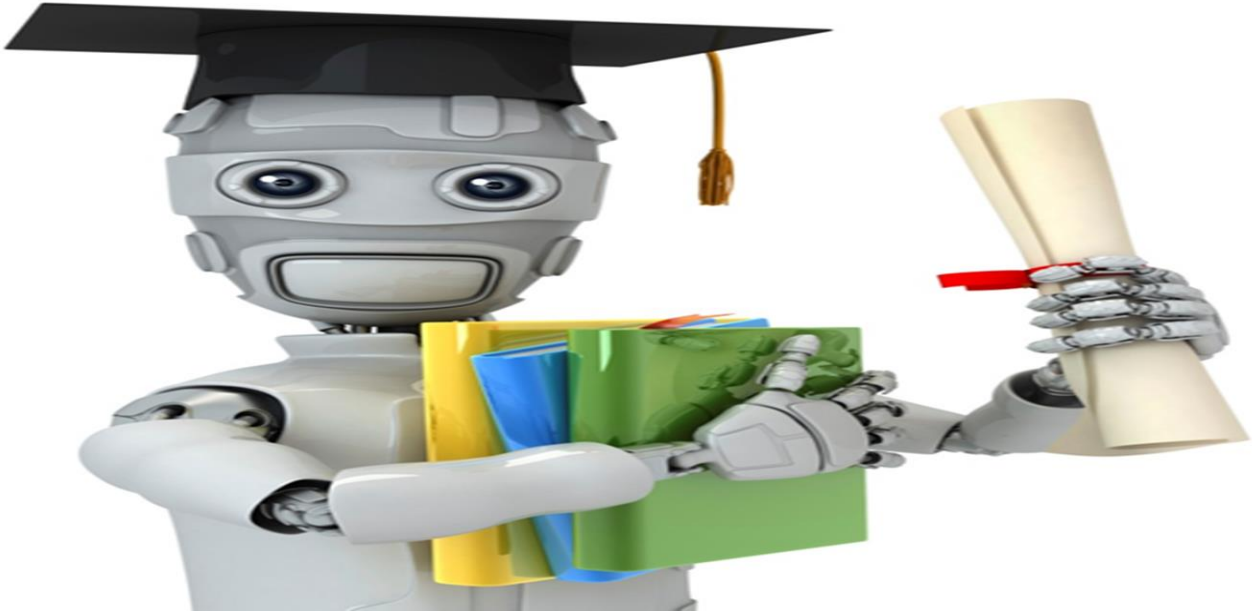
Lecture 1:
Introduction to Artificial Intelligence

CSC 411

Artificial Intelligence



Introduction - Required Reading

Akinsola, J. E. T., Adeagbo, M. A., Oladapo, K., Akinsehinde, S. A., & Onipede, F. O. (2022). Artificial Intelligence Emergence in Disruptive Technology. In A. O. Salau, S. Jain, & M. Sood (Eds.), *Computational Intelligence and Data Sciences: Paradigms in Biomedical Engineering*. CRC Press, Taylor & Francis Group, U.K.

COURSE OUTLINE

- Overview/Introduction
- Intelligent Agents
- Problem Solving
- Knowledge and Reasoning
- Application Areas in AI
- Seminar Presentations

Introduction - Required Reading

- International grandmaster Arnold Denker studies the pieces on the board in front of him. He realizes there is no hope; he must resign the game. His opponent, HITECH, becomes the first computer program to defeat a grandmaster in a game of chess (Berliner, 1989).

Introduction - Required Reading

- "I want to go from Boston to San Francisco," the traveller says into the microphone. "What
date will you be travelling on?" is the reply. The traveller explains she wants to go October 20th,
nonstop, on the cheapest available fare, returning on Sunday. A speech understanding program
named PEGASUS handles the whole transaction, which results in a confirmed reservation that
saves the traveller $894 over the regular coach fare. Even though the speech recognizer gets one
out of ten words wrong, it is able to recover from these errors because of its understanding of
how dialogs are put together (Zue et al., 1994).

Introduction - Required Reading

- An analyst in the Mission Operations room of the Jet Propulsion Laboratory suddenly starts paying attention. A red message has flashed onto the screen indicating an "anomaly" with the Voyager spacecraft, which is somewhere in the vicinity of Neptune. Fortunately, the analyst is able to correct the problem from the ground. Operations personnel believe the problem might have been overlooked had it not been for MARVEL, a real-time expert system that monitors the massive stream of data transmitted by the spacecraft, handling routine tasks and alerting the analysts to more serious problems (Schwuttke, 1992).

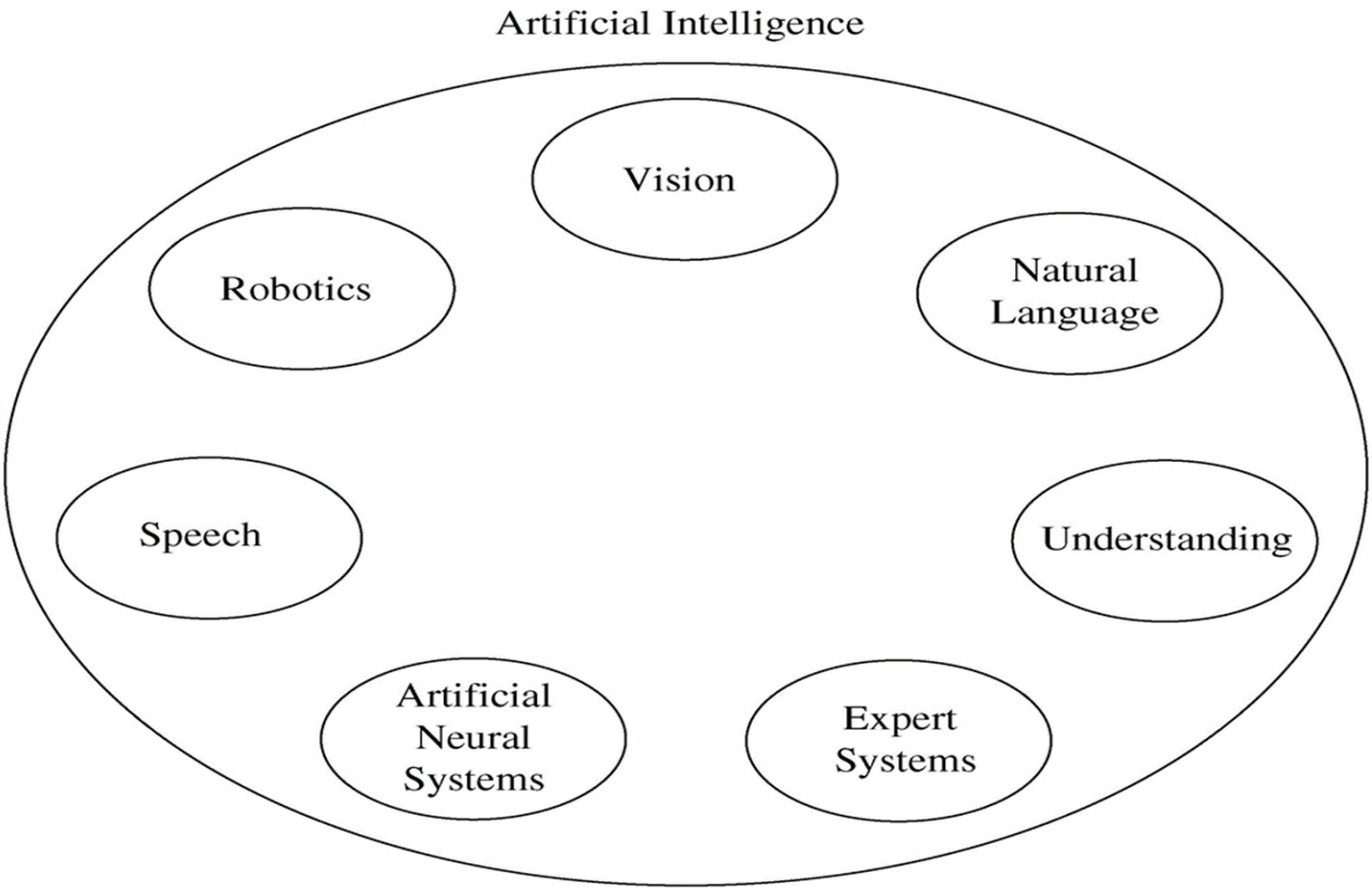
Introduction - Required Reading

- Cruising the highway outside of Pittsburgh at a comfortable 55 mph, the man in the driver's
seat seems relaxed. He should be—for the past 90 miles, he has not had to touch the steering wheel, brake, or accelerator. The real driver is a robotic system that gathers input from video cam eras, sonar, and laser range finders attached to the van. It com bines these inputs with experience learned from training runs an d successfully computes how to steer the vehicle (Pomerleau, 1 993).

According to Herbert Simon (1957):

- It is not my aim to surprise or shock you – but the simplest way I can summarize is to say that there are now in the world machines that think, that learn and that create. Moreover, their ability to do these things is going to increase rapidly until…in the visible future…the range of problems they can handle will be coextensive with the range to which human mind has been applied.

SOME AREAS/SUBSETS OF AI



Overview

- What is intelligence?
- What is Artificial Intelligence?
- How can intelligence be measured?
- How does our brain work?
- Intelligent machine?
- Science fiction?
- Rebuild human mind?

What is intelligence?

- What can you deduce from the following statements (they sound axiomatic based on popular usage…but are they truly so?):
  - Cleanliness is next to godliness…
  - Heaven helps those who help themselves…
  - Hard work begets success…
  - The Devil you know is better than the Angel you don't know…
- FOLKS! HOW COMMON IS COMMON SENSE?

What is intelligence?

- Try precisely defining intelligence. It is next to impossible. Despite the wide use (and misuse) of terms such as intelligent systems, there is no widely agreed-upon scientific definition of intelligence. It is therefore useful to think of intelligence in terms of an open collection of attributes.

- Intelligent entities
  - Perceive
  - Reason
  - Choose
  - Act
  - Achieve goals
  - Learn
  - Reflect
  - Communicate
  - Organize

What is intelligence?

- Perception: manipulation, integration, and interpretation of data provided by sensors
- Reasoning: deductive (logical) inference, inductive inference, analogical inference | including reasoning in the face of uncertainty and incomplete information, hypothetical reasoning, justification and explanation of inferences, evaluation of explanations, adapting explanations in the light of falsified assumptions or changing world states.
- Action: coordination, control, and use of effectors/actuators to accomplish a variety of tasks including exploration and manipulation of the environment, including design and construction of tools towards this end.

What is intelligence?

- Adaptation and Learning: adapting behaviour to better cope with changing environmental demands, discovery of regularities, explanation of observations in terms of known facts and hypotheses, construction of task-specific internal representations of the environment, discovery of procedures, learning to differentiate despite similarities and generalize despite differences, learning to describe specific domains in terms of abstract theories and concepts, learning to use, adapt, and extend language, learning to reason, plan, and act.

What is intelligence?

- Reflection and Awareness: of internal processes (e.g., reasoning, goals, etc.) of self as well as other agents.
- Communication: with other intelligent agents including humans using signals, signs, icons, symbols, sound, pictures, touch, language and other communication media | including communication of goals, desires, beliefs, narratives of real and imaginary episodes, explanation of actions and events.
- Organization: into social groups based on shared objectives, development of shared conventions to facilitate orderly interaction, culture.

What is intelligence?

- Autonomy: Setting of goals, deciding on the appropriate course of actions to take in order to accomplish the goals or directives (without explicit instructions from another entity), executing the actions to satisfy the goals, adapting the actions and/or goals as necessary to deal with any unforeseen circumstances (to the extent permitted by the agent's physical capabilities and the environmental constraints).
- Creativity: exploration, modification, and extension of domains (e.g., language, mathematics, music) by manipulation of domain-specific constraints, or by other means.

What is Artificial Intelligence?

AI is about:

- ❑ Study of computational models of intelligence
- ❑ Falsifiable hypotheses about intelligent behavior
- ❑ Construction of intelligent artifacts (agents)
- ❑ Mechanization of tasks requiring intelligence
- ❑ Exploring the design space of intelligent systems

What is Artificial Intelligence?

- Different people think of AI differently. Two important questions to ask are:
- Are you concerned with thinking or behavior?
- Do you want to model humans, or work from an ideal standard?

Definitions of AI

SYSTEMS THAT ACT LIKE HUMANS	SYSTEMS THAT ACT RATIONALLY
SYSTEMS THAT THINK LIKE HUMANS	SYSTEMS THAT THINK RATIONALLY

Definitions of AI

- Bellman (1978)
- The automation of activities that we associate with human thinking activities such as decision making, problem solving, learning etc.
- Haugeland (1985)
- The exciting new effort to make computers think…machines with minds, in the full and literal sense.
- Charniak and McDermott (1985)
- The study of mental faculties through the use of computational models
- Kurzweil (1990)
- The art of creating machines that perform functions that require intelligence when performed by humans

Definitions of AI

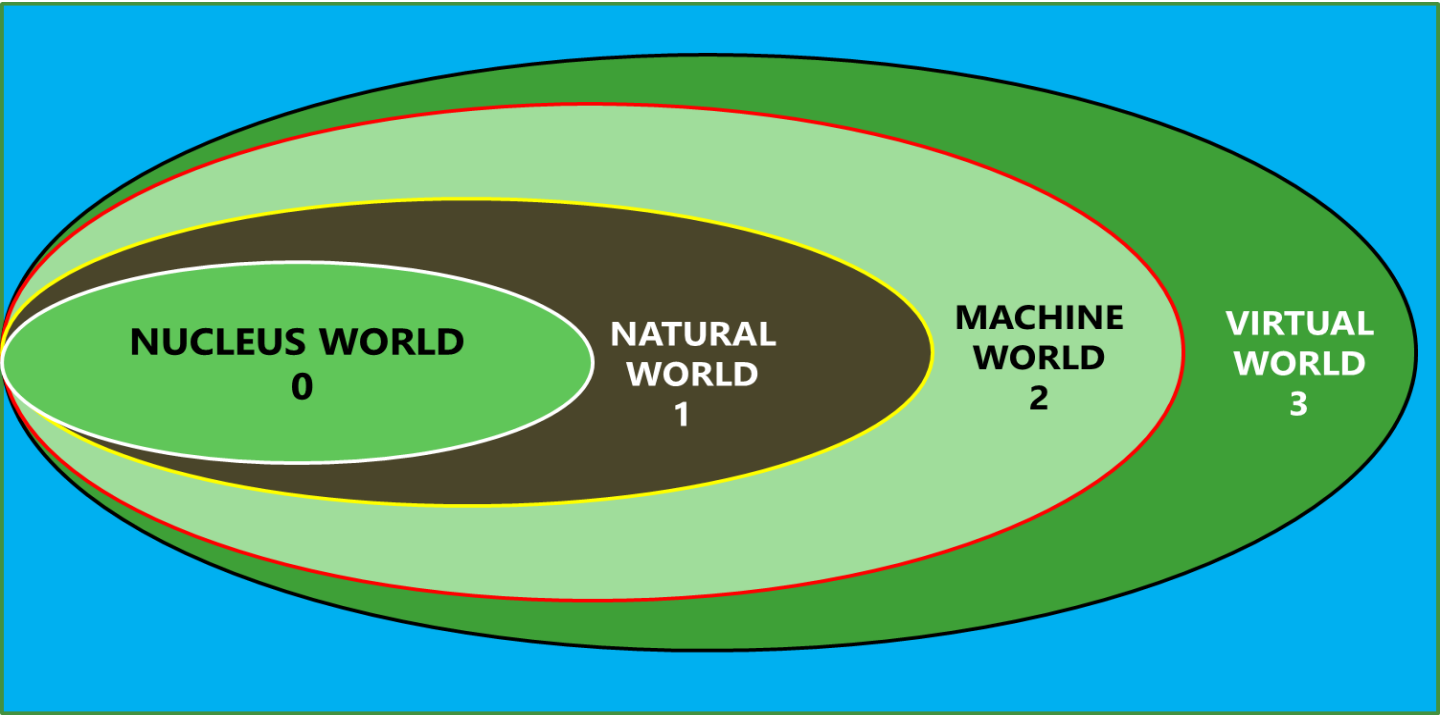
- Schalkoff (1990)
- A field of study that seeks to explain and emulate intelligent behavior in terms of computational processes
- Winston (1992)
- The study of the computations that make it possible to perceive, reason and act.
- John McCarthy (1995)
- The aim of AI is to develop machines that behave as if they were intelligent
- Stubblefield (2003)
- The branch of Computer Science that is concerned with the automation of intelligence

Definitions of AI

- Encyclopedia Britannica:
  - AI is the ability of a digital computer or computer-controlled robot to perform tasks commonly associated with intelligent beings.
  - According to this definition, every computer is and AI system.
- Elaine Rich:
  - AI is the study of how to make computers do things at which, at the moment, people are better.
  - Still up-to-date in the year 2050!
  - Humans are still better in many fields (e.g. understanding pictures, learning ability)!
  - Computers are already better in many fields (e.g. chess playing)!

Modern View: Virtual Machine (1)

- Each definition we have examined so far depends on the author's world view and scope of work. In defining AI from the view of making intelligent agents, a 4-world model is used



0	NUCLEUS WORLD	This is the world from which all other world s originate
1	NATURAL WORLD	This is referred to as the natural or intellige nt world. It comes as a result of the activiti es of World 0
2	MACHINE WORLD	This is the machine or abstract world. Just as World 1 came as a result of the activitie s of World 0, World 2 comes as a result of t he activities of World 1
3	VIRTUAL WORLD	This is referred to as the symbolic or virtual world

Generally speaking, what happens in world “i + 1” comes as a resul t of the happenings in world i.

Foundations of AI

- Although AI is can be said to be a relatively young field, it inherited many ideas, viewpoints and techniques from other disciplines:
- Philosophy: (going back to 400 B.C.) made AI conceivable by considering the ideas that the mind is in some ways like a machine, that it operates on knowledge encoded in some internal language, and that thought can be used to help arrive at the right actions to take.
- Mathematics: Mathematicians provided the tools to manipulate statements of logical certainty as well as uncertain, probabilistic statements. They also set the groundwork for reasoning about algorithms.

Foundations of AI

- Psychology: Psychologists strengthened the idea that humans and other animals can be considered information processing machines.
- Linguistics: Linguists showed that language use fits into this model as proposed by the psychologists.
- Computer Engineering: Computer engineering provided the artifact that makes AI applications possible. AI programs tend to be large, and they could not work without the great advances in speed and memory that the computer industry has provided.

The Road to Artificial Intelligence

- History of computer science is really a history of human
- attempts to understand nous (rational mind)
- 1931 – The Austrian Kurt Godel shows that in first-order predicate logic all true statements are derivable. In higher-order logics, on the other hand, there are true statements that are unprovable.
- 1937 – Alan Turing points out the limits of intelligent machines with the halting problem.
- 1943 – McCulloch and Pitts model neural networks and make the connection to propositional logic.
- 1950 – Alaln Turing defines machine intelligence with the Turing Test and writes about learning machines and genetic algorithms.
- 1951 – Marvin Minsky and Dean Edmonds develops a neural network machine. With 3000 vacuum tubes he simulates 40 neurons.
- 1955 – Arthur Samuel (IBM) builds a learning checkers program that plays better than its developer.

The road to Artificial Intelligence

- 1956 – McCarthy organizes a conference in Dartmouth College. Here the name Artificial Intelligence was first introduced.
- Newell and Simon of Carnegie Mellon University (CMU) present the Logic Theorist (LT), the first symbol-processing computer program.
- 1958 – McCarthy invents at MIT (Massachusetts Institute of Technology) the high-level language LISP. He writes programs that are capable of modifying themselves.
- 1959 – Gelernter (IBM) builds the Geometry Theorem Prover.
- 1961 – The General Problem Solver (GPS) by Newell and Simon imitates human thought.
- 1963 – McCarthy founds the AI Lab at Stanford University.
- 1965 – Robinson invents the RESOLUTION CALCULUS for predicate logic (??).
- 1966 – Weizenbaum's program ELIZA carries out dialogue with people in natural language.

The road to Artificial Intelligence

- 1969 – Minsky and Papert show in their book Perceptrons that the perceptron, a very simple neural network, can only represent linear functions.
- 1972 – French Scientist Alain Colmerauer invents the logic programming language PROLOG.
- British physician de Dombal develops an expert system for diagnosis of acute abdominal pain. It goes unnoticed in the mainstream AI community of the time.
- 1976 – Shortliffe and Buchanan develop MYCIN, an expert system for diagnosis of infectious diseases, which is capable of dealing with uncertainty.
- 1981 – Japan begins, at great expense, the "Fifth Generation Project" (a 10-year plan) with the goal of building intelligent computers running PROLOG in much the same way that ordinary computers run machine code.
- 1982 – R1, the expert system for configuring computers, saves Digital Equipment Corporation (DEC) 40million dollars per year

The road to Artificial Intelligence

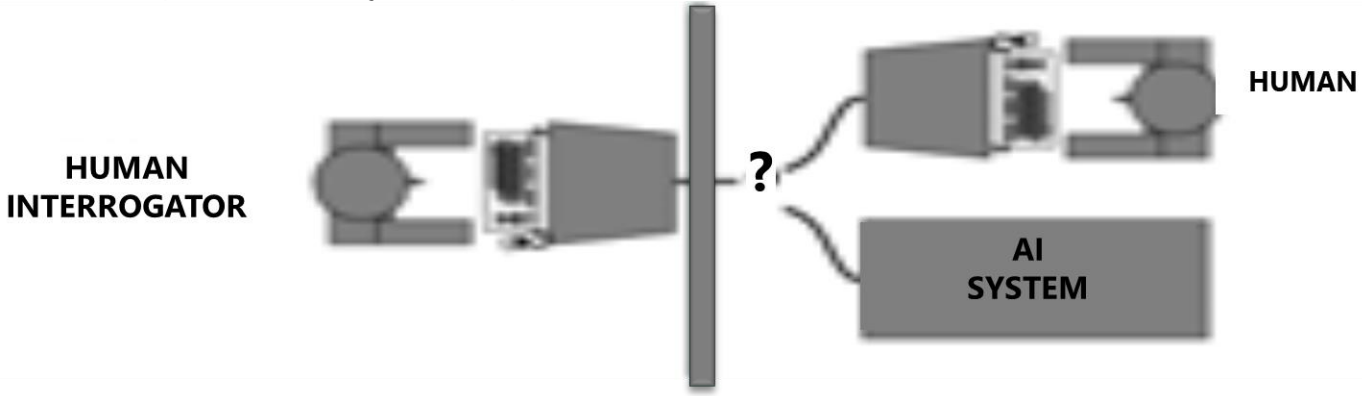
- 1986 – Renaissance of neural networks through, among others, Rumelhart, Hinton and Sejnowski. The system Nettalk learns to read texts aloud.
- 1990 – Pearl, Cheeseman, Whittaker, Spiegelhalter bring probability theory into AI with Bayesian networks.
- Multi-agent systems became popular.
- 1992 – Tesauros TD-gammon program demonstrates the advantages of reinforcement learning.
- 1993 – Worldwide RoboCup initiative to build soccer-playing autonomous robots.
- 1995 – From statistical learning theory, Vapnik develops support vector machines, which are very imoportant today.
- 1997 – First international RoboCup competition in Japan.
- 2003 – The robots in RoboCup demonstrate impressively what AI and robotics are capable of achieving.
- 2006 – Service robotics becomes a major AI research area.
- 2010 – Autonomous robots start learning their policies.

Why AI?

- "AI can have two purposes. One is to use the power of compute rs to augment human thinking, just as we use motors to augmen t human or horse power. Robotics and expert systems are major branches of that. The other is to use a computer's artificial intelli gence to understand how humans think. In a humanoid way. If you test your programs not merely by what they can accomplis h, but how they accomplish it, they you're really doing cognitive science; you're using AI to understand the human mind."
- - Herb Simon

How can intelligence be measured?

- The Turing Test – A measure of AI's success?
- – Requires all major components of AI:



- Knowledge representation: system is able to comprehend the information provided before, during and after an interrogation
- Automated reasoning: using stored information to make decisions
- Natural language processing: allows for the system to communicate using any human language e.g. English
- Machine learning: allows systems t detect new patterns

How does our brain work?

- Massive research works are ongoing in different areas of AI. There are still a great deal of challenges seeking breakthrough. Take for instance some of the early difficulties encountered in the subset of Natural Language processing/translation.

- A translation was made from Russian to English that should mean:
- "the spirit is willing, but the flesh is weak" as "the vodka is good but the meat is rotten"

Intelligent machine?

- What is your take on the claim "computers are not intelligent, they can only do what their programmers tell them"
- What about science fiction?
- Can the human mind be rebuilt/replicated in machines?

Thank you!